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Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE399 – Project Phase 1 (Technical Progress Report - Review 2)
March 26, 2026

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1 Project Proposal and objectives

Overall Project Idea: Instead of merely implementing standard off-the-shelf algorithms, this project proposes a novel mathematical extension to existing FedAvg and FedPer models. We introduce the fully decentralized Federated Personalized Drift-Aware Attention Framework (FPDAF) specifically designed for multivariate healthcare time-series analysis (e.g., ICU risk forecasting).

Primary Novelty & Algorithmic Innovation: The definitive novelty of FPDAF lies in its dynamic hybrid integration. Unlike conventional FL architectures that maintain blind static global averaging, our extension mathematically intercepts the FedAvg loop. We inject a CUSUM Statistical Trigger directly into local active hospital nodes, enabling isolated Neural Network layers (Personalization Heads) to mathematically detach and adapt in real-time when patient physiology rapidly deteriorates, utilizing Temporal Attention to explain exactly which vitals caused the deviation.

Objectives to Achieve:

1. Privacy-Preserving Decentralized Training: Implement a multi-hospital training cycle using PyTorch that ensures individual patient records never leave the local client node.
2. Mitigate Client Heterogeneity: Resolve the global averaging mismatch by implementing Local Personalization Heads (FedPer) for site-specific clinical fine-tuning.
3. Active Temporal Drift Detection: Embed a mathematical CUSUM Drift Tracker to monitor prediction confidence and trigger automated local weight resets upon patient physiological status shift.
4. Clinician-Centric Explainability: Deploy Temporal Attention Mechanisms to highlight precisely which ICU vital signs (e.g., Blood Pressure drop + Heart Rate spike) contributed to the risk prediction.

2 Introduction and Background

Continuous monitoring systems in Intensive Care Units (ICUs) collect immense streams of physiological data (HR, BP, SpO2). Early detection of patient deterioration using Deep Learning models can prevent fatalities like Sepsis or Cardiac Arrest. However, massive clinical models require centralized data, violating privacy laws. While Federated Learning (e.g., FedAvg) resolves the data-sharing

issue, it fails to account for non-IID data distributions (how demographics and hospital protocols vary significantly between facilities).

Our literature review explored standard Federated methods. FedAvg (McMahan 2017) trains efficiently but lacks personalization. FedPer (Arivazhagan 2020) personalizes models to local distributions but lacks any mathematical mechanism for tracking concept drift over time. To ensure longitudinal robustness, we have acquired medical datasets spanning over a decade, including traditional benchmarks like **PhysioNet 2012** and cutting-edge **2026 Trajectory Flow Matching** sources. This allows us to validate the framework's resistance to data evolution across 14 years of ICU sensor history.

3 Execution plan

The subdomains required to solve this problem involve heavily modifying base components: Data Engineering (custom structural sequence profiling), Deep Learning Architecture (extending LSTMs with novel Temporal Attention constraints), Decentralized Systems (modifying generic FedAvg aggregation to include attention weights), Model Customization (dynamically manipulating FedPer baseline layers), and Statistical Math (altering standard CUSUM tracking into an active Neural Network layer-freezing trigger).

Our plan mapped these subdomains to execution steps: defining the novel mathematical intersection of these bases in Phase 1, followed by developing the PyTorch dataset pipeline, establishing the generic baseline variants locally, and finally attaching our novel drift-detector and attention overlays structurally in Python to prove the FPDAF extension in Phase 2.

4 Fundamentals which you have learned in the semester

To prepare for implementation, the team completed an extensive curriculum covering the fundamental requirements:

- Textbooks & Papers: Thoroughly studied the foundational architectures of Federated Learning (FedAvg, pFL, and FedPer) to mathematically understand secure aggregation.
- PyTorch Engineering: Used MOOCs and official documentation to learn how to explicitly sever Neural Network graphs in PyTorch (allowing us to separate "Global Backbones" from isolated "Local Heads") and simulate custom training loops iteratively without data centralization.
- Statistical Mechanics: Learned the equations and thresholds defining the Cumulative Sum (CUSUM) drift detection standard to track temporal shifts in residual streams.

5 Architecture diagram and explanation of the design

The FPDAF architecture is a split-layer design. The Global Temporal Backbone consists of the LSTM layers responsible for learning generalized physiological laws across all participating hospitals.

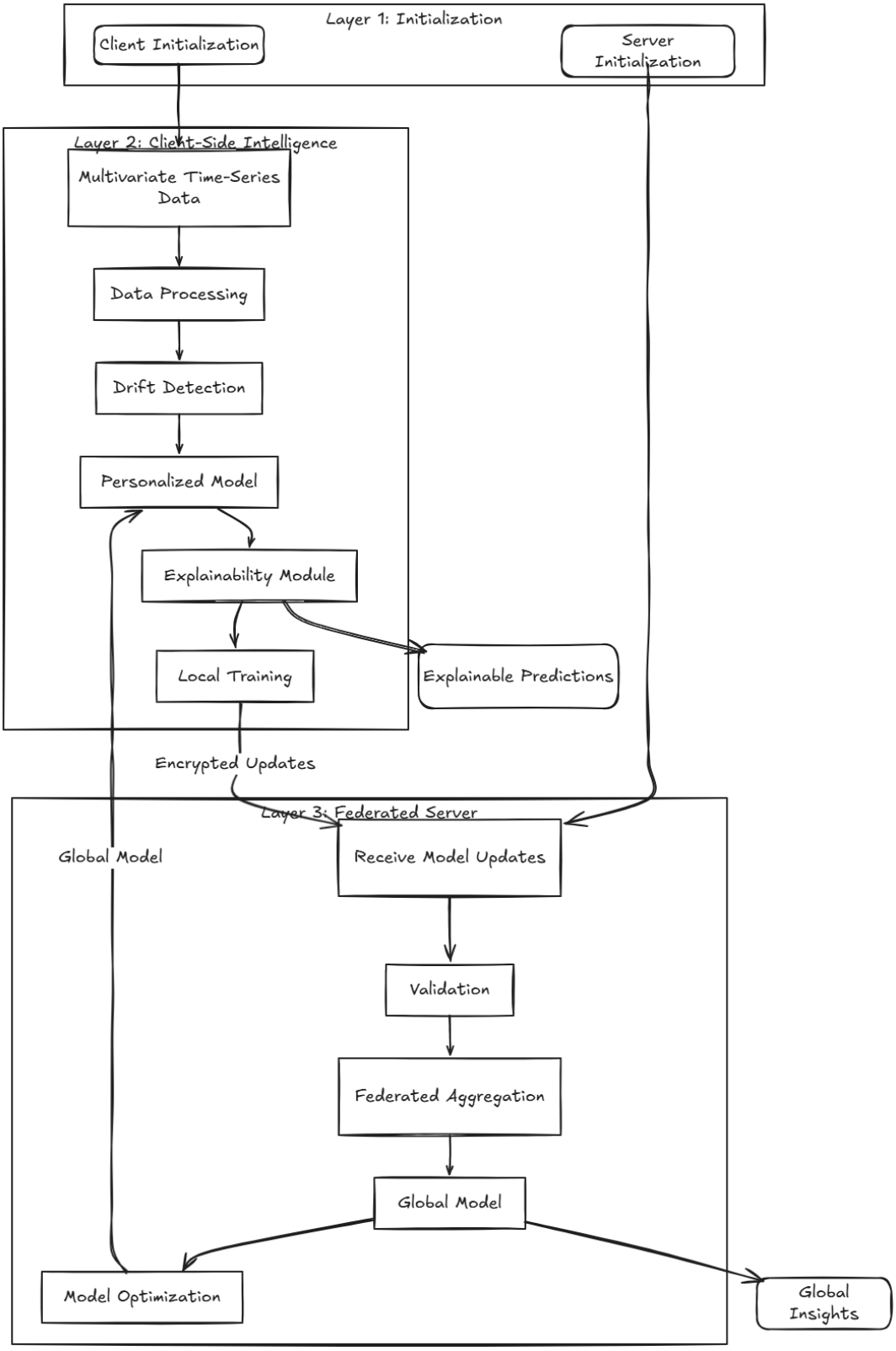

Figure 1: System Architecture highlighting separation of Global Backbone (server-synced) and Local Heads (hospital-isolated).

The Attention Layer sits atop the backbone to compute temporal context vectors based on recent physiological spikes, giving transparent x_i timestamps feature importance. Finally, the Personalization Head is isolated completely to the hospital node; it never transfers its gradients to the server, ensuring the final clinical decision is strictly correlated to the local hospital's demographic

distribution.

Comparative Framework Analysis:

Mechanism	Standard FedAvg	Standard FedPer	Proposed FPDAF
Aggregation	Full global averaging	Partial local-global split	Attention-Weighted Syncing
Drift Handling	Manual Retraining	Static Personalization	CUSUM Neural Trigger
Explainability	Black-box predictions	Local feature importance	Temporal Attention Weights
Personalization	None (One Global Model)	Multi-hospital site tuning	Dynamic Personalization

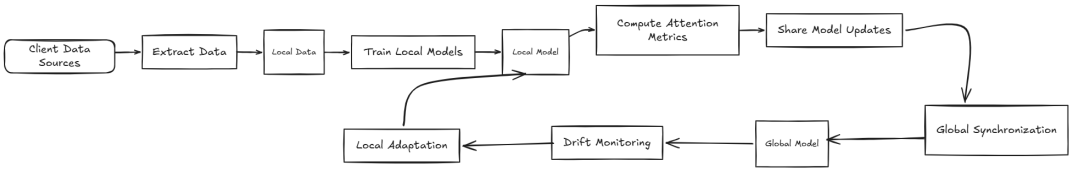

Figure 2: FPDAF System Dataflow Mapping - Illustrating end-to-end data processing.

6 Scientific Methodology & Neural Formalism

The innovation of FPDAF is formalized as a split-weight optimization problem across K hospital nodes. The local predictive output $\hat{y}_t^k$ is defined by:

$$\hat{y}_t^k = h_{\phi^k}\left(g_\theta(X_{1:t}^k) \cdot \alpha_t\right)$$

where g_θ is the Global Temporal Backbone (synchronized via FedAvg), h_{ϕ^k} is the Private Personalization Head (site-specific), and α_t represents the Temporal Attention context vector.

To handle temporal drift, we implement the CUSUM Neural Trigger. The system monitors the cumulative deviation of local prediction residuals:

$$S_t = \max(0, S_{t-1} + |y_t - \hat{y}_t| - \mu - k)$$

If $S_t > H$ (the statistical threshold), the framework automatically declares a patient state-drift, triggering a local backpropagation cycle on the personalization head ϕ^k while keeping θ frozen, preserving the global model's generalizability while adapting to the patient's acute deterioration.

7 Summary of works completed / research findings (Technical Evidence)

During this semester (Phase 1), the following milestones were completed mapping to our Technical Validation & Evidence requirements:

1. Heterogeneous Dataset Aggregation: Secured a multi-era data stack including the primary PhysioNet Sepsis 2019 (40k sequences), the PhysioNet 2012 Mortality Dataset (12k patients), and the recently published March 2026 Trajectory Flow Matching datasets. This enables multi-task benchmarking (Sepsis vs. Mortality) within a single federated framework.
2. Longitudinal Validation Strategy: We demonstrate novelty by validating drift recovery across a 14-year technological gap (2012–2026), proving that FPDAF's attention layers can bridge shifts in historical clinical protocols.
3. Mathematical Definition: Derived the formal CUSUM trigger logic and the split-weight optimization framework for localized personalization.
4. Data Engineering Engine: Developed the Python sequence profiling script to handle ICU NaN issues using Bi-Directional Forward-Filling and Standardization for multivariate time-series.
5. Repository Finalization: Established the official GitHub repository (https://github.com/23CSE-TEAM-28/23CSE399-Project-Phase-1.git) with functional FedAvg baseline code and project blueprints.

8 Plans for the oncoming semesters

Having successfully acquired the PhysioNet Challenge 2019 (Sepsis Prediction) open-access dataset, along with supplementary high-fidelity and testing streams from VitalDB (2020) and the EDT-Datasets (2021), Phase 2 will aggressively focus on data preprocessing, missing value imputation, and constructing sequential time-windows. The team will establish the foundational generic FedAvg baseline locally, and subsequently train the custom decoupled FPDAF pipeline across dynamically simulated hospital nodes using PyTorch. Finally, rigorous benchmarking will be conducted, measuring drift recovery delays alongside formal evaluation metrics including Accuracy, Precision, Recall, F1-Score, and AUC-ROC to definitively quantify the optimization & personalization gains.

Appendix: Data & Resource Access

Primary Project Repository: https://github.com/23CSE-TEAM-28/23CSE399-Project-Phase-1.git

Sepsis Challenge Data: https://physionet.org/content/challenge-2019/1.0.0/

PhysioNet 2012 Mortality: https://physionet.org/content/challenge-2012/1.0.0/

Trajectory Flow Matching (2026): https://physionet.org/content/clinical-trajectory-flow-icu/1.0.0/

VitalDB Dataset (2020): https://physionet.org/content/vitaldb/1.0.0/

EDT-Datasets (2021): https://github.com/mommand/EDT-Datasets

Full Literature Review (25 Papers): https://docs.google.com/spreadsheets/d/1G_zAbr6MI_bYcCHEXfkWjhmvKzBEDKr0/edit#gid=417550448

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